

Notes: Hong, Kubik, and Stein (JF, 2004)

Social Interaction and Stock-Market Participation

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1 Big picture

- **Question.** Does social interaction make households more likely to participate in the stock market?
- **Setting.** The paper uses the 1992 Health and Retirement Study (HRS), a survey of about 7,500 U.S. households with detailed information on stockholding, wealth, education, demographics, and several direct proxies for sociability. A smaller 1998 “war-baby” HRS cohort is used for an out-of-sample check.
- **Core challenge.** Households that know neighbors or attend church may differ from less social households for many reasons unrelated to peer effects: wealth, education, race, optimism, openness to new experiences, or risk tolerance. The empirical challenge is to distinguish social interaction from these latent personality traits.
- **Main conceptual contribution.** The paper shifts attention from standard fixed-cost explanations of participation to a *social* fixed-cost story. Social households face lower effective participation costs when more of their peers already invest.
- **Main hypotheses.**
 - **Higher participation for socials:** households with more social interaction should be more likely to own stock, conditional on wealth and demographics.
 - **Social multipliers:** the participation effect of sociability should be stronger where participation is already high, because more local investors means more opportunities for information sharing or social reinforcement.
 - **Possible multiple equilibria:** if peer effects are strong enough, both low-participation and high-participation local equilibria can be self-sustaining.
- **Main empirical design.** The authors run household-level linear probability regressions of stock ownership on three sociability proxies—knowing neighbors, visiting neighbors, and attending church—while controlling for wealth, education, race, age, urban residence, marital status, risk tolerance, and additional personality proxies.
- **Main baseline result.** Social households are more likely to own stocks. Depending on the sociability measure and specification, the marginal effect is about 2–4 percentage points, with the strongest estimates around 4 percentage points for knowing neighbors or attending church.
- **Key heterogeneity result.** The social effect is much larger for households that are white/non-Hispanic, educated, and wealthier, and it is stronger in states where stock market participation is higher.
- **Key contrast result.** The interaction of sociability with state participation is present, but the interaction of *risk tolerance* with state participation is not. This is important because risk tolerance should matter for participation but not through social channels.
- **Main credibility result.** The sociability coefficients are robust to many controls for wealth, income, risk tolerance, pessimism, technological openness, homeownership, household composition, health, and state fixed effects. They also do not predict checking-account ownership, which weakens a generalized “trust in financial institutions” interpretation.

- **Economic takeaway.** Social interaction appears to lower the effective cost of entering the stock market and can amplify participation booms. The paper provides a microfoundation for why stock ownership can diffuse socially over time.

2 Data and measurement (key objects)

2.1 Why the Health and Retirement Study is a useful laboratory

- The HRS first surveyed households in 1992, focusing on families with at least one member born between 1931 and 1941. The average age in the first wave is about 56.
- The survey includes detailed information on asset holdings, demographics, education, wealth, marital status, and several direct questions about social interaction.
- Later waves in 1994, 1996, and 1998 follow the same households. The paper mostly uses the 1992 wave because later waves add little independent information and attrition is somewhat selective.
- In 1998 the HRS added a smaller “war-baby” cohort, born between 1942 and 1947, with roughly 1,400 households. This sample is used only for a limited robustness exercise because it is much smaller and omits several key questions.

Interpretation. The HRS is valuable because it combines stockholding behavior with unusually direct survey measures of how socially connected households are.

2.2 Sample construction and household definitions

- The paper uses 7,465 household observations from the 1992 HRS.
- When age and education are reported for both spouses, household age and years of education are taken as the *higher* of the two values.
- A household is classified as non-white if either member is non-white.
- The main sample is restricted to the original 1992 survey. The 1998 war-baby sample is examined separately in one comparison table.

Interpretation. The unit of analysis is the household, not the individual, and the authors intentionally define household characteristics in a way that captures the strongest available human-capital margin in the family.

2.3 Defining stock-market participation

The dependent variable in the main regressions is a dummy equal to one if the household owns stock and zero otherwise.

Important details:

- The stockholding measure includes direct stock ownership and mutual funds.

- It covers *non-retirement-account* assets only, because the HRS contains very little information on retirement-account equity.
- In the 1992 HRS sample, 26.7% of households own stock under this definition.

Interpretation. The paper studies a broad equity-participation decision, but only outside retirement accounts. Measured participation rates are therefore lower than in surveys that include 401(k)s and IRAs.

2.4 Social-interaction variables

The paper uses three binary indicators of sociability.

- **Know Neighbors.** From the question “Of your closest neighbors, how many do you know?” The dummy equals one if the answer is *all*, *most*, or *some*, and zero if the answer is *none*. In the sample, 92.9% of households are coded social by this measure.
- **Visit Neighbors.** From the question “How often do you visit with your neighbors?” The dummy equals one for daily, several times a week, several times a month, or several times a year, and zero for hardly ever. The sample mean is 63.9%.
- **Attend Church.** From the question “How often do you attend religious services?” The dummy equals one if the household attends at least once a year and zero if it never attends. The sample mean is 76.0%.

Only one household member—typically the woman—answers these social-interaction questions.

Interpretation. These variables are not peer *actions* like observed stockholding of friends. They are direct proxies for the intensity of social contact through which information or reinforcement may flow.

2.5 Personality proxies and other controls

To address the concern that sociability proxies may simply proxy for personality, the paper adds several further controls:

- **Risk Tolerant.** Equals one if at least one household member says yes to a Barsky-style job gamble with a 50–50 chance of doubling income or cutting it by one-third. Mean: 32.5%.
- **Depressed.** Equals one if at least one household member reports feeling depressed “all or almost all of the time” during the past week. Mean: 8.3%.
- **Low Tech.** Equals one if both household members answer “don’t do” to a question about using a computer or word processor. Mean: 30.9%.
- Demographic controls include years of education, age, white/non-Hispanic indicator, urban indicator, marital-status dummies, and wealth controls.
- In richer specifications, the authors also add income dummies, education dummies, age dummies, state dummies, and the depression and low-tech indicators.

Interpretation. **Risk Tolerant** is the cleanest non-social personality proxy. **Depressed** and **Low Tech** may themselves partly reflect social connectedness, so they are useful controls but not perfect placebo variables.

2.6 State-level participation measures

To test social-multiplier implications, the paper classifies states according to stock market participation in two ways:

- **Raw participation rate:** the fraction of households in the state that own stock.
- **Abnormal participation rate:** the average residual from a household participation regression that controls for wealth, education, race, age, urban status, and marital status.

The state-level indicator is coded as -1 , 0 , or 1 depending on whether a state is in the low-, medium-, or high-participation group.

Interpretation. The abnormal participation measure is the paper's preferred proxy because it strips out broad compositional differences across states and is closer to a reduced-form measure of participation costs or participation culture.

2.7 Descriptive facts

- Overall stock ownership is 26.7%.
- Participation increases sharply with wealth: from 3.41% in the bottom wealth quintile to 55.03% in the top quintile.
- Participation differs strongly by race and education: 34.28% for white/non-Hispanic households versus 9.21% for non-white/Hispanic households; 49.53% for college graduates versus 19.97% for non-college households.
- The raw social differences are also large. For *Know Neighbors*, participation is 27.79% for social households and 12.33% for non-social households. For *Visit Neighbors*, the corresponding figures are 28.92% and 22.77%. For *Attend Church*, they are 27.70% and 23.52%.
- The strongest pairwise correlation among the sociability measures is between *Know Neighbors* and *Visit Neighbors* (0.37). Both are only weakly correlated with *Attend Church* (about 0.08–0.09).
- *Risk Tolerant* is essentially uncorrelated with the social variables, while *Depressed* and *Low Tech* are modestly negatively correlated with them.

Interpretation. The raw data already suggest a strong role for sociability, but they also show why aggressive controls are needed: wealth, race, and education all matter enormously for participation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper’s published JF version sketches the theoretical model verbally; a more formal treatment appears in Hong, Kubik, and Stein (2001). The core idea can be summarized as follows.

For a *non-social* investor i :

$$\text{Participate if } B_i(\text{wealth, risk, education}) - F_i \geq 0, \quad (1)$$

where F_i is a fixed participation cost that does not depend on others.

For a *social* investor in peer group g :

$$\text{Participate if } B_i(\cdot) - (F_i - \phi P_g) \geq 0, \quad \phi > 0, \quad (2)$$

where P_g is peer-group participation.

Interpretation. Social interaction lowers the effective fixed cost of participation. This may work through word-of-mouth learning about returns and trading technology, or through utility from discussing the market with other participants.

The model yields three predictions:

- social households should participate more;
- participation of social households should respond more strongly to exogenous shocks such as lower market access costs;
- sufficiently strong social effects can generate multiple equilibria in participation.

3.2 Baseline cross-sectional specification

The main empirical model is a household-level linear probability regression:

$$\Pr(\text{OwnStock}_i = 1) = \alpha + \beta S_i + X_i' \gamma + \varepsilon_i, \quad (3)$$

where:

- OwnStock_i is a dummy for stock market participation,
- S_i is one of the three sociability indicators,
- and X_i includes wealth, education, age, race, urban status, marital status, and personality controls.

There are three baseline variants:

- a parsimonious specification with four wealth-quintile dummies;
- a richer wealth-control specification with 19 wealth-bin dummies;
- a fully saturated version that also adds income dummies, state dummies, age dummies, education dummies, depression, and low-tech controls.

Interpretation. The coefficient β measures the partial association between sociability and stock ownership after conditioning on observed household characteristics.

3.3 Estimation choice

- All main regressions are estimated by OLS as linear probability models.
- Because the dependent variable is binary, the authors also re-estimate the models with probit and logit. The results are very similar.
- The data are cross-sectional, not panel. Identification therefore comes from rich conditioning, not from within-household changes over time.

Interpretation. The paper values transparency of coefficients in percentage-point terms. The LPM makes the magnitudes especially easy to read.

3.4 Heterogeneity by demographic group

To test whether sociability matters most for households near the participation margin, the paper estimates:

$$\Pr(\text{OwnStock}_i = 1) = \alpha + \beta S_i + \delta(S_i \times H_i) + X_i' \gamma + \varepsilon_i, \quad (4)$$

where H_i is an indicator for being white/non-Hispanic, educated, and wealthy.

In Table IV, “educated” means that at least one household member has completed high school, and “wealthy” means wealth above the sample median.

Interpretation. If social interaction reduces effective fixed costs, it should matter more for households close to entering the market and less for groups far below the participation threshold.

3.5 State-level interaction specification

The paper’s strongest credibility exercise augments the baseline with state-level participation interactions:

$$\Pr(\text{OwnStock}_i = 1) = \alpha + \beta S_i + \delta(S_i \times P_s) + \lambda R_i + \mu(R_i \times P_s) + X_i' \gamma + \varepsilon_i, \quad (5)$$

where:

- $P_s \in \{-1, 0, 1\}$ classifies the household’s state into low, medium, or high participation,
- R_i is the *Risk Tolerant* dummy,
- and X_i includes the same demographic controls as in the richer baseline.

The paper runs this with both *raw* and *abnormal* state participation.

Interpretation. This is the key design that separates social interaction from generic personality. Sociability should matter more in high-participation states if peer effects are real; risk tolerance should not.

3.6 Intertemporal comparison using the 1998 war-baby sample

The authors compare the baseline coefficient on *Visit Neighbors* between the 1992 HRS and the 1998 war-baby cohort:

$$\Pr(\text{OwnStock}_{it} = 1) = \alpha_t + \beta_t \text{VisitNeighbors}_{it} + X'_{it}\gamma_t + \varepsilon_{it}, \quad t \in \{1992, 1998\}. \quad (6)$$

The 1998 sample excludes the risk-tolerance control because that variable is unavailable there.

Interpretation. This is a tentative test of whether the social-participation gradient became stronger during the stock-market boom of the 1990s.

3.7 Checking-account placebo specification

To test whether the social variables simply proxy for generalized trust in financial institutions, the paper runs:

$$\Pr(\text{CheckingAccount}_i = 1) = \alpha + \beta S_i + X'_i\gamma + \varepsilon_i, \quad (7)$$

with the same covariates as in the baseline participation regressions.

Interpretation. If the social variables just measure broad trust, they should help explain checking-account ownership as well as stock ownership. If they do not, the stock-market result is more specific.

3.8 Standard errors and robustness

- The paper reports robust standard errors throughout.
- Additional robustness exercises redefine participation using minimum stockholding thresholds of \$2,500, \$5,000, \$10,000, and \$25,000; the results are very similar.
- Further controls for vacations, household size, employment status, health, and homeownership do not materially change the sociability coefficients.

Interpretation. The paper's basic participation result is not driven by households with trivial stock positions or by omitted proxies for how busy or socially available households are.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no conventional instrument

The paper does *not* use an instrumental variable. Identification is based on conditional comparisons across households after a rich set of controls.

Interpretation. The paper is transparent that its evidence is not experimental. Its credibility comes from the pattern of results across a sequence of sharper tests rather than from a single exclusion restriction.

4.2 Main identification concern: endogenous sociability

The core concern is that households who know neighbors or attend church may differ in unobserved ways that also affect stockholding.

Examples:

- they may be more risk tolerant;
- they may be more optimistic about returns;
- they may be more open to new experiences;
- or they may simply have more free time to process financial information.

The baseline regressions therefore control for risk tolerance, depression, low-tech status, wealth, income, age, education, race, marital status, and urban residence.

Interpretation. The initial strategy is “selection on observables,” with unusually rich personality and demographic controls for this literature.

4.3 Why the state-level interaction is the key credibility design

The paper’s strongest argument is that if the sociability variables were merely proxies for something like risk tolerance, then their effect should not systematically vary with the state-level participation rate.

Instead, the paper finds:

- sociability matters much more in high-participation states;
- the same is *not* true for risk tolerance.

Interpretation. This is the most convincing evidence in the paper because it introduces a prediction unique to the peer-effects interpretation.

4.4 What the paper can rule out

The paper provides evidence against several alternative interpretations:

- **Pure demographics.** Wealth, race, and education are tightly controlled.
- **Generalized trust in financial institutions.** Social variables do not predict checking-account ownership.
- **Simple cross-state omitted variables.** Mean state participation by itself predicts ownership, but the authors argue that this is harder to interpret than the state-level interaction design.

Interpretation. The paper can rule out a large class of simple compositional or trust-based stories.

4.5 What the paper cannot fully distinguish

- The data do not let the authors separate cleanly between **word-of-mouth learning** and **utility from discussing the market with others**. Both are consistent with the main results.

- The paper cannot distinguish whether socials invest more because they have *more interactions* or because they are *better listeners* who learn more from a given level of interaction.
- The paper therefore does not claim the kind of causal interpretation one would get from randomized assignment of peers or forced social exposure.

Interpretation. The paper is very strong on economic patterning and comparative statics, but more modest on deep mechanism and causal policy interpretation.

4.6 Relation to the literature

The paper connects three literatures:

- stock market participation and fixed costs (e.g., Vissing-Jorgensen),
- peer effects and social interactions,
- and social multipliers in participation decisions.

Its distinctive contribution is to bring *direct measures of sociability* into the participation literature and to show that peer effects appear stronger exactly where the theory says they should be stronger.

5 Tables and figures

5.1 Table I: stock-market participation rates for different household categories

Read:

- Panel A shows that stock ownership rises sharply with wealth, education, and being white/non-Hispanic.
- Overall participation is 26.70%; it rises from 3.41% in the lowest wealth quintile to 55.03% in the highest.
- Panel B gives the raw social comparison. Using *Know Neighbors*, participation is 27.79% for social households and 12.33% for non-social households.
- With *Visit Neighbors*, the gap is smaller but still positive: 28.92% versus 22.77%.
- With *Attend Church*, participation is 27.70% for social households and 23.52% for non-social households.

Economic message. The raw patterns already point toward a strong social gradient in participation, but they also show why wealth and education must be controlled aggressively.

Table I
Stock-Market Participation Rates for Different Categories
of Households

The entries are the stock-market participation rates for different categories of households in the Health and Retirement Study conducted in 1992. Stock-market participation rates pertain only to non-retirement-account assets. Panel A reports these rates by race, education, and wealth of households. Household wealth is defined as the value of all assets excluding non-retirement stock-holdings. Panel B reports these rates by three household sociability measures (Know Neighbors, Visit Neighbors, and Attend Church) and wealth. For Know Neighbors, households who say they do not know neighbors are Not Social; the rest are Social. For Visit Neighbors, households who say they do not visit neighbors are Not Social; the rest are Social. For Attend Church, households who say they never attend church are Not Social; the rest are Social. There are 7,465 household observations.

Panel A: Stock-Market Participation Rates by Race, Education, and Wealth					
	All (1)	Racial Groups		Education Levels	
		White and Non-Hispanic (2)	Non-White or Hispanic (3)	College Graduate (4)	No College Degree (5)
All Households	26.70%	34.28%	9.21%	49.53%	19.97%
1 st Quintile of wealth distribution	3.41%	6.39%	1.17%	6.36%	3.17%
2 nd Quintile of wealth distribution	11.34%	15.28%	5.37%	27.01%	9.28%
3 rd Quintile of wealth distribution	24.10%	27.81%	13.32%	36.36%	20.91%
4 th Quintile of wealth distribution	39.84%	43.73%	21.32%	50.11%	35.42%
5 th Quintile of wealth distribution	55.03%	57.45%	35.93%	68.38%	44.39%

Panel B: Stock-Market Participation Rates by Sociability and Wealth						
	Know Neighbors		Visit Neighbors		Attend Church	
	Not Social (1)	Social (2)	Not Social (3)	Social (4)	Not Social (5)	Social (6)
All Households	12.33%	27.79%	22.77%	28.92%	23.52%	27.70%
1 st Quintile of wealth distribution	2.05%	3.67%	3.63%	3.22%	4.05%	3.16%
2 nd Quintile of wealth distribution	11.40%	11.34%	11.46%	11.27%	12.24%	11.02%
3 rd Quintile of wealth distribution	14.29%	24.52%	22.20%	25.05%	23.59%	24.25%
4 th Quintile of wealth distribution	30.91%	40.18%	37.03%	41.17%	37.14%	40.56%
5 th Quintile of wealth distribution	41.18%	55.52%	50.86%	56.92%	50.49%	56.21%

Table II
Means and Correlations of Sociability Measures and Other Independent Variables

The entries are summary statistics on various characteristics of households in the Health and Retirement Study conducted in 1992. Know Neighbors equals zero for the households who say they do not know any neighbors, and one otherwise. Visit Neighbors equals zero for the households who say they do not visit neighbors, and one otherwise. Attend Church equals zero for the households who say they never attend church, and one otherwise. Additional household characteristics include indicator variables for whether one of the respondents is risk tolerant (Risk Tolerant), for whether the household has one member who suffers from depression (Depressed), and for whether the household avoids using computers or word processors (Low Tech). Other characteristics include an indicator for whether the household's reported race is White and Non-Hispanic, Years of Education (of the most educated person in the household), and Wealth (household wealth excluding non-retirement stockholdings). There are 7,465 household observations.

	Know Neighbors (1)	Visit Neighbors (2)	Attend Church (3)	Risk Tolerant (4)	Depressed (5)	Low Tech (6)	White Non-Hispanic (7)	Years of Education (8)	Wealth (9)
Mean of sample	92.94%	63.87%	76.02%	32.53%	8.33%	30.92%	69.75%	12.65	\$196,405
Correlations:									
Know Neighbors	1.000								
Visit Neighbors	0.3665	1.000							
Attend Church	0.0792	0.0942	1.000						
Risk Tolerant	-0.0118	-0.0213	0.0035	1.000					
Depressed	-0.0683	-0.0689	-0.0305	-0.0034	1.000				
Low Tech	-0.0589	-0.0342	-0.0228	0.0505	0.0920	1.00			
White Non-Hispanic	0.0872	0.0311	-0.1472	-0.0016	-0.0917	-0.2050	1.00		
Years of Education	0.0832	0.0585	0.0171	0.0345	-0.1587	-0.3785	0.3226	1.00	
Wealth	0.0537	0.0352	0.0310	0.0114	-0.0641	-0.0900	0.1481	0.1845	1.00

5.2 Table II: means and correlations of sociability measures and controls

Read:

- The mean of *Know Neighbors* is 92.94%, of *Visit Neighbors* 63.87%, and of *Attend Church* 76.02%.
- *Know Neighbors* and *Visit Neighbors* are fairly strongly correlated (0.3665), while both are only weakly correlated with church attendance.
- *Risk Tolerant* is almost uncorrelated with the social measures.
- *Depressed* and *Low Tech* are negatively correlated with sociability, especially with knowing neighbors.

Economic message. The table suggests that the three social variables overlap but are not identical, and that risk tolerance is empirically distinct from sociability.

5.3 Table III: baseline effect of sociability on whether a household owns stocks

Table III
Baseline Effect of Sociability on Whether a Household Owns Stocks

The sample comprises households in the Health and Retirement Study conducted in 1992. The dependent variable is an indicator that the household owns stock. The independent variables are: a Sociability indicator (either Know Neighbors, Visit Neighbors, or Attend Church), Risk Tolerant indicator, Years of Education, Age, White/Non-Hispanic indicator, an Urban indicator, Depressed indicator, Low Tech indicator, marital status dummies (not shown), and a sex-of-respondent dummy (not shown). The specifications in columns (1), (4), and (7) include dummies for wealth quintiles. The others include dummies for every 5 percent of the wealth distribution, and in some cases, dummies for every year of age and education, for every 5 percent of the income distribution and for each state. Robust standard errors are in parentheses. There are 7,465 observations. (Numbers with * are significant at 10-percent level. Numbers with ** are significant at 5-percent level. Numbers with *** are significant at 1-percent level.)

	Know Neighbors			Visit Neighbors		Attend Church		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sociability indicator	0.0411*** (0.0141)	0.0406*** (0.0142)	0.0405*** (0.0141)	0.0215** (0.0092)	0.0218*** (0.0092)	0.0247*** (0.0092)	0.0369*** (0.0109)	0.0355*** (0.0109)
Risk Tolerant indicator	0.0367*** (0.0099)	0.0369*** (0.0099)	0.0311*** (0.0099)	0.0369*** (0.0099)	0.0371*** (0.0099)	0.0314*** (0.0099)	0.0364*** (0.0099)	0.0366*** (0.0099)
Years of Education	0.0221*** (0.0015)	0.0220*** (0.0015)	0.0221*** (0.0015)	0.0229*** (0.0015)	0.0229*** (0.0015)	0.0229*** (0.0015)	0.0229*** (0.0015)	0.0228*** (0.0015)
Age	0.0005 (0.0010)	0.0005 (0.0010)	0.0005 (0.0010)	0.0005 (0.0010)	0.0005 (0.0010)	0.0005 (0.0010)	0.0005 (0.0010)	0.0005 (0.0010)
White/Non-Hispanic indicator	0.0995*** (0.0096)	0.1003*** (0.0096)	0.0990*** (0.0105)	0.0998*** (0.0096)	0.1003*** (0.0097)	0.0988*** (0.0105)	0.1069*** (0.0099)	0.1075*** (0.0100)
								0.1044*** (0.0108)

(continued)

(continued)

Table III—Continued

	Know Neighbors			Visit Neighbors			Attend Church		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Urban indicator	0.0605*** (0.0166)	0.0622*** (0.0166)	0.0481*** (0.0168)	0.0509*** (0.0166)	0.0616*** (0.0166)	0.0477*** (0.0168)	0.0600*** (0.0166)	0.0616*** (0.0166)	0.0476*** (0.0168)
Depressed indicator			-0.0302** (0.0126)			-0.0292** (0.0127)			-0.0306** (0.0126)
Low Tech indicator			-0.0349*** (0.0098)			-0.0352*** (0.0098)			-0.0350*** (0.0098)
2 nd Quintile of wealth distribution	0.0281*** (0.0099)			0.0298*** (0.0099)			0.0302*** (0.0099)		
3 rd Quintile of wealth distribution	0.1156*** (0.0120)			0.1177*** (0.0120)			0.1176*** (0.0120)		
4 th Quintile of wealth distribution	0.2455*** (0.0149)			0.2476*** (0.0149)			0.2462*** (0.0149)		
5 th Quintile of wealth distribution	0.3757*** (0.0160)			0.3776*** (0.0159)			0.3760*** (0.0160)		
19 Wealth dummies	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Age dummies	No	No	Yes	No	No	Yes	No	No	Yes
Years of Education dummies	No	No	Yes	No	No	Yes	No	No	Yes
19 Income dummies	No	No	Yes	No	No	Yes	No	No	Yes
State dummies	No	No	Yes	No	No	Yes	No	No	Yes

Read:

- The baseline coefficients on the sociability indicator are positive in every specification.
- For *Know Neighbors*, the coefficient ranges from about 0.0405 to 0.0622 across specifications.
- For *Visit Neighbors*, it ranges from about 0.0215 to 0.0247.
- For *Attend Church*, it ranges from about 0.0268 to 0.0369.
- These estimates survive the addition of finer wealth controls, income controls, state dummies, depression, and low-tech controls.

Economic message. Social interaction is robustly associated with higher stock market participation, even after conditioning on the standard drivers of entry and several personality proxies.

5.4 Table IV: interactions of sociability with wealth, education, and race

Read:

- The interaction between sociability and the white/educated/wealthy indicator is large and positive for *Visit Neighbors* and *Attend Church*.
- For *Know Neighbors*, the baseline social coefficient is 0.0254 and the interaction is 0.0546, implying an effect of roughly 8 percentage points for the high-participation demographic group.
- For *Visit Neighbors*, the implied effect among white, educated, wealthy households is about 0.059.
- For *Attend Church*, the implied effect is about 0.071.

Economic message. The social effect is much larger for households closer to the participation margin and in demographic groups where participation is already common.

Table IV Interactions of Sociability with Wealth, Education, and Race			
The sample comprises households in the Health and Retirement Study conducted in 1992. The dependent variable is an indicator that the household owns stock. The independent variables are the Sociability indicator (either Know Neighbors, Visit Neighbors, or Attend Church) and the Sociability indicator interacted with the White, Educated, and Wealthy indicator. The White, Educated, and Wealthy indicator equals one for households who are white/non-Hispanic, who have at least one high-school graduate and who have wealth above the sample median. The regression specifications also include all of the control variables used in column (2) of Table III (Risk Tolerant indicator, Years of Education, Age, White and Non-Hispanic indicator, Urban indicator, dummies for every 5 percent of the wealth distribution, marital status dummies, and a sex-of-resident dummy). There are 7,465 household observations. Numbers with * are significant at 10-percent level. Numbers with ** are significant at 5-percent level. Numbers with *** are significant at 1-percent level.			
	Know Neighbors (1)	Visit Neighbors (2)	Attend Church (3)
Sociability indicator	0.0254** (0.0127)	-0.0034 (0.0080)	0.0038 (0.0114)
Sociability indicator * White, Educated, and Wealthy indicator	0.0546 (0.0519)	0.0628** (0.0215)	0.0672** (0.0240)

Table V Low, Medium, and High Participation States		
The sample comprises households in the Health and Retirement Study conducted in 1992. Panel A reports raw participation rates by state. A state's raw participation rate is the fraction of households in the state that own stock. Panel B reports abnormal participation rates by state. A state's abnormal participation rate is its average residual in a regression of participation against all of the controls in column (1) of Table III except for the Sociability indicator (Risk Tolerant indicator, Years of Education, Age, White and Non-Hispanic indicator, Urban indicator, dummies for every quintile of the wealth distribution, marital status dummies, and a sex-of-resident dummy).		
Low Participation States	Medium Participation States	High Participation States
Panel A: Raw Participation Rates		
Alabama (14.81%)	South Carolina (23.91%)	Michigan (31.31%)
Louisiana (12.38%)	Florida (24.01%)	Maryland (32.89%)
Arkansas (15.59%)	Virginia (25.83%)	New Jersey (35.45%)
Tennessee (17.27%)	Colorado (25.97%)	Indiana (34.49%)
North Dakota (17.39%)	Illinois (28.51%)	Massachusetts (34.79%)
Texas (18.04%)	Oklahoma (28.57%)	New Hampshire (35.85%)
North Carolina (18.57%)	Illinois (28.56%)	Nevada (37.55%)
Mississippi (18.58%)	Oregon (29.09%)	Kansas (39.02%)
Georgia (19.44%)	Massachusetts (29.25%)	Wisconsin (39.82%)
District of Columbia (21.73%)	Ohio (29.34%)	Connecticut (40.30%)
New York (21.78%)	Iowa (30.06%)	Washington (40.36%)
Arizona (23.48%)	Pennsylvania (31.11%)	Maine (44.22%)
West Virginia (23.68%)		Wyoming (45.83%)
Panel B: Abnormal Participation Rates (Residual Relative to Median State: Iowa)		
North Dakota (-17.27%)	Florida (-1.70%)	South Carolina (2.76%)
Massachusetts (-7.72%)	New York (-1.52%)	Maryland (3.12%)
Tennessee (-7.24%)	Illinois (-0.42%)	Pennsylvania (3.14%)
Arkansas (-7.11%)	Virginia (-0.17%)	Massachusetts (3.35%)
Louisiana (-6.43%)	Oklahoma (-0.11%)	Michigan (3.63%)
Alabama (-6.20%)	Iowa (0%)	Kansas (3.92%)
New Hampshire (-5.39%)	Washington (0%)	Ohio (4.19%)
Oregon (-4.67%)	North Carolina (0.45%)	Connecticut (5.21%)
Mississippi (-4.13%)	New Jersey (1.45%)	Nebraska (5.79%)
Arizona (-2.02%)	Georgia (2.02%)	Indiana (6.18%)
Colorado (-1.87%)	District of Columbia (2.03%)	Wisconsin (6.23%)
California (-1.85%)	Texas (2.15%)	Maine (6.29%)
West Virginia (-1.77%)		Wyoming (15.21%)

Table VI Interactions of Sociability and Risk Tolerance with State-level Participation Rates					
The sample comprises households in the Health and Retirement Study conducted in 1992. The dependent variable is an indicator that the household owns stock. The independent variables are the Sociability indicator (either Know Neighbors, Visit Neighbors, or Attend Church) and the Sociability indicator interacted with the State-level Participation indicator. The State-level Participation indicator takes on the values -1, 0, or 1 depending on whether the household's state has low, medium, or high stock-market participation. In column (1), (3), and (5), state-level participation is measured on a raw basis. In column (2), (4), and (6), it is measured on an abnormal basis. The other independent variables shown in the table are the Risk Tolerant indicator and its interaction with the State-Level Participation indicator. The regression specifications also include Years of Education, Age, a White and Non-Hispanic indicator, an Urban indicator, dummies for every 5 percent of the wealth distribution, a sex-of-resident dummy, marital status dummies, and the State-Level Participation indicator. Robust standard errors are in parentheses. There are 7,465 household observations. Numbers with * are significant at 10-percent level. Numbers with ** are significant at 5-percent level. Numbers with *** are significant at 1-percent level.					
	Know Neighbors		Visit Neighbors		Attend Church
	Raw (1)	Abnormal (2)	Raw (3)	Abnormal (4)	Raw (5)
Sociability indicator	0.0474** (0.0145)	0.0425** (0.0145)	0.0245** (0.0095)	0.0218** (0.0095)	0.0373** (0.0110)
Sociability * state-level participation indicator	0.0460** (0.0285)	0.0471** (0.0287)	0.0063 (0.0222)	0.0114 (0.0228)	0.0085 (0.0141)
Risk Tolerant indicator	0.0349** (0.0150)	0.0362** (0.0150)	0.0352** (0.0100)	0.0367** (0.0100)	0.0345** (0.0100)
Risk Tolerant indicator * state-level participation indicator	-0.0117 (0.0129)	-0.0146 (0.0134)	-0.0109 (0.0129)	-0.0139 (0.0130)	-0.0118 (0.0135)

5.5 Tables V and VI: state-level participation and social multipliers

Read:

- Table V classifies states into low, medium, and high participation groups using both raw and abnormal participation rates.

- Table VI then interacts sociability with that state-level participation classification.
- Using the preferred abnormal-participation measure, the interaction is positive in all three cases and strongly significant for *Know Neighbors* and *Attend Church*.
- In the *Know Neighbors* regression, the difference between social and non-social households is roughly -0.5 percentage points in low-abnormal-participation states but rises to about 9 percentage points in high-participation states.
- By contrast, the interaction between *Risk Tolerant* and state participation is small and insignificant in every specification.

Economic message. This is the strongest evidence for a peer-effects interpretation. Sociability matters more where there are more investors to interact with, but risk tolerance does not.

5.6 Table VII: comparing the 1992 and 1998 samples

Read:

- The coefficient on *Visit Neighbors* is 0.0204 in the 1992 sample and 0.0351 in the 1998 war-baby sample.
- The point estimate therefore rises by roughly 75%.
- However, the 1998 estimate is imprecise because the sample is much smaller and excludes some controls, so equality with the 1992 coefficient cannot be rejected at conventional levels.

Economic message. The data are suggestive of an intertemporal social multiplier during the 1990s boom, but the paper is appropriately cautious because the test has limited power.

Table VII
Baseline Effect of Sociability on Whether a Household Owns Stocks
Using the 1992 and 1998 Samples

The samples comprise households in the Health and Retirement Study surveys conducted in 1992 and in 1998. The dependent variable is an indicator that the household owns stock. The Sociability indicator is the Visit Neighbors measure. The regression specifications also include Years of Education, Age, a White and Non-Hispanic indicator, an Urban indicator, dummies for every quintile of the wealth distribution, marital status dummies, and a sex-of-respondent dummy. Robust standard errors are in parentheses. There are 7,465 households in the 1992 sample and 1,441 households in the 1998 sample. (Numbers with * are significant at 10-percent level. Numbers with ** are significant at 5-percent level. Numbers with *** are significant at 1-percent level.)

	1992 Sample (1)	1998 Sample (2)
Sociability indicator	0.0204** (0.0093)	0.0351 (0.0233)
Years of Education	0.0239*** (0.0015)	0.0238*** (0.0045)
Age	0.0004 (0.0010)	0.0030 (0.0036)
White/Non-Hispanic indicator	0.0956*** (0.0094)	0.0769*** (0.0260)
2 nd Quintile of wealth distribution	0.0282*** (0.0099)	0.0812*** (0.0353)
3 rd Quintile of wealth distribution	0.1133*** (0.0129)	0.1429*** (0.0369)
4 th Quintile of wealth distribution	0.2449*** (0.0149)	0.2971*** (0.0383)
5 th Quintile of wealth distribution	0.3726*** (0.0159)	0.4676*** (0.0394)

Table VIII
The Effect of Sociability on Whether a Household Has
a Checking Account

The sample comprises households in the Health and Retirement Study conducted in 1992. The dependent variable is an indicator that the household has a checking account. The independent variables are the Sociability indicator (either Know Neighbors, Visit Neighbors, or Attend Church), Risk Tolerant indicator, Years of Education, Age, a White and Non-Hispanic indicator, and an Urban indicator. The regression specifications also include dummies for every 5 percent of the wealth distribution, marital status dummies, and a sex-of-respondent dummy. Robust standard errors are in parentheses. There are 7,465 household observations, of which 78.22 percent have checking accounts. (Numbers with * are significant at 10-percent level. Numbers with ** are significant at 5-percent level. Numbers with *** are significant at 1-percent level.)

	Know Neighbors (1)	Visit Neighbors (2)	Attend Church (3)
Sociability indicator	-0.0116 (0.0175)	-0.0062 (0.0081)	0.0132 (0.0092)
Risk Tolerant indicator	-0.0247*** (0.0084)	-0.0248*** (0.0084)	-0.0246*** (0.0084)
Years of Education	0.0235*** (0.0016)	0.0235*** (0.0016)	0.0234*** (0.0016)
Age	0.0006 (0.0008)	0.0006 (0.0008)	0.0005 (0.0008)
White/Non-Hispanic indicator	0.1357*** (0.0109)	0.1356*** (0.0109)	0.1380*** (0.0110)
Urban indicator	0.0341** (0.0141)	0.0343** (0.0141)	0.0346** (0.0141)

5.7 Table VIII: checking-account placebo

Read:

- Replacing stock ownership with checking-account ownership as the dependent variable makes the sociability coefficients small and statistically insignificant.

- The coefficients are actually negative for two of the three social proxies.
- *Risk Tolerant* enters strongly negatively, and the white/non-Hispanic indicator enters strongly positively.

Economic message. The social variables do not appear to proxy for a generalized trust in financial institutions. Their explanatory power is specific to stock market participation.

6 Short summary

- The paper studies whether social interaction raises stock market participation.
- It uses three direct sociability proxies from the HRS: knowing neighbors, visiting neighbors, and attending church.
- Social households are 2–4 percentage points more likely to own stock in baseline regressions.
- The effect is much larger for white, educated, and wealthier households and in high-participation states.
- The strongest credibility result is that sociability interacts with state-level participation, but risk tolerance does not.
- The checking-account placebo shows that sociability is not simply proxying for broad trust in financial institutions.
- The paper cannot fully separate word-of-mouth learning from utility derived from discussing the market, but it provides strong evidence that social interaction lowers the effective cost of entering the stock market.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that more social households are more likely to participate in the stock market and that this relationship is strongest exactly where peer-based mechanisms should matter most: in demographic groups with high participation and in states where many others already invest.

7.2 Economic intuition

Stock market participation is not just an individual cost-benefit calculation. Social interaction can reduce effective fixed costs by spreading information, by making participation more salient, or by making the market a topic of shared conversation. Because these effects are stronger where more peers are already in the market, participation can exhibit social multipliers.

7.3 Takeaway

The paper's lasting message is that peer interaction helps households cross the participation threshold. That makes stock market participation partly a *social diffusion process*, not only an individual portfolio choice.